



DRIFTGUARD ENTERPRISE

• STANDARD OPERATING PROCEDURE (SOP) | B2B CLIENT RETAINER SERVICE RUNBOOK



• ARBITRUM NITRO CONSENSUS

DriftGuard

Sub-130ms Failover Gateway & Active-Passive Sentinel

<130ms Failover · 250ms Tip Sync · 42MB RAM · Zero Drops

B2B Retainer Service Operations Runbook

Turnkey Ingress Orchestration, Out-of-Band Consensus Sentinel Monitoring & 24/7 Incident Escalation for Arbitrum Orbit L3s, Game Studios & Account Abstraction Bundlers

DOCUMENT IDENTIFIER

DG-SOP-B2B-001 (Version 2.0)

TARGET CLIENT BASE

Orbit L3 Rollups | Game Studios | ERC-4337 Bundlers

GATEWAY AVAILABILITY SLA

99.99% Availability Guarantee (Contractual Credits)

KEY CUSTODY POLICY

Zero Custody (No Private Keys or Signers Handled)

COMMERCIAL PRICING TIERS

\$2,500 / \$5,000 / \$8,500 per Month

INCIDENT RESPONSE SLA

< 15 min Sev-1 Escalation (24/7 Paging)

MAINTAINER CORE TEAM

DriftGuard Systems Infrastructure Group

SECURITY CLASSIFICATION

INTERNAL MAINTAINER & CLIENT OPERATIONS

Commercial Framework Intent: This Standard Operating Procedure governs the commercial retainer operations of DriftGuard Systems. It establishes the client onboarding protocol, sidecar injection patterns, drift calibration tables, 24/7 severity response SLAs, and monthly upstream vendor reporting delivered under paid B2B enterprise service agreements.

1. Executive Overview & Service Catalog

The **DriftGuard Enterprise Resilience Retainer** is a managed B2B infrastructure service providing turnkey L7 ingress management, out-of-band consensus verification, 24/7 incident response, and upstream provider resilience guarantees.

While the core DriftGuard software daemon remains 100% open-source and MIT-licensed, enterprise Web3 organizations require guaranteed uptime SLAs, dedicated on-call engineers, custom upstream pool calibration, and continuous chaos resilience auditing without hiring in-house distributed systems specialists.

Target Client Profiles & Acute Pain Points

Client Segment	Critical Pain Point	DriftGuard Retainer Solution	Value Delivered
Arbitrum Orbit L3 Appchains (Gaming, Social, DeFi)	Sequencer feed disconnects or AnyTrust DAC storage desyncs stall the entire chain.	Custom raw WebSocket sequencer listener + dedicated multi-node gateway cluster.	Guarantees continuous sequencer availability and sub-50ms disconnect mitigation.
ERC-4337 Paymaster Bundlers (Biconomy, ZeroDev, Pimlico)	Lagging RPC nodes return stale nonces, causing paymaster transactions to revert with nonce too low.	Monotonically guaranteed nonce routing and sub-130ms socket drain away from lagging nodes.	Eliminates failed user onboarding and relayer queue locks.
DeFi Liquidators & Relayers (GMX, Camelot, Pendle Keepers)	A 3-second RPC delay (12 missed Nitro blocks) results in missed liquidations and arbitrage losses.	Real-time consensus drift sentinel auditing blocks every 200ms out-of-band.	Protects MEV opportunities and prevents stale oracle contract queries.
Dedicated Web3 Game Studios (Real-Time Action Games)	Stale RPC reads cause "ghost items", inventory desync, and player session disconnects.	Sub-130ms transparent cutover without player TCP disconnection or game client code changes.	Preserves player experience and continuous server-authoritative synchronization.

2. Retainer Tiers & Pricing Schedule

Retainer Tier	Monthly Price	Target Organization	Service Level Agreement & Deliverables
Tier 1: Sentinel Standard	\$2,500 / month	dApps, Indexers & Relayers	Up to 3 chain backends; 24/7 automated Discord/Telegram webhook alerts; monthly upstream provider health audit; 2-hour business-hours incident triage response.
Tier 2: Orbit Rollup Mission-Critical	\$5,000 / month	Orbit L3 Chains & Game Studios	Dedicated multi-node HA cluster sidecar architecture; custom consensus anchors (client validator + 2 private tier-1 RPCs); < 15 min Sev-1 incident SLA with 24/7 on-call paging; bi-weekly synthetic chaos injection drills; post-mortems within 24 hours.
Tier 3: Institutional Sovereign / Paymaster	\$8,500 / month	High-Volume Paymasters & Sovereign Chains	Active-active multi-region gateway orchestration; custom WebSocket raw sequencer feed listener; AnyTrust Data Availability Committee (DAC) custom sentry; dedicated senior systems engineer on-call; custom Grafana enterprise dashboard; 99.99% availability guarantee .

3. SOP-01: Client Onboarding & Pre-Flight Audit

Objective: Audit the client's current RPC architecture, quantify peak throughput, identify single-points-of-failure (SPOFs), and establish baseline latency metrics within 5 business days.

5-Day Pre-Flight Audit Workflow

- Day 1 (Discovery & Inventory Intake):**
 - Document Chain ID(s), upstream RPC providers (Alchemy, Infura, QuickNode, local validator), and peak queries-per-second (QPS).
 - Classify client consumer types: ERC-4337 paymaster bundler, Go/Rust relay daemon, game server backend, or public dApp frontend.
- Day 2 (Upstream Provider Resilience Profiling):**
 - Benchmark client's current upstream providers over 48 hours for p50/p90/p99 response latency, block propagation lag against the canonical sequencer, and HTTP 429 rate-limiting frequency.
- Day 3 (Consensus Anchor Selection):**
 - Select 2 independent, non-correlated reference nodes (e.g., direct sequencer endpoint + private tier-1 fallback like dRPC/Tenderly) to ensure out-of-band monitoring integrity.
- Day 4-5 (Pre-Flight Audit Report Delivery):**
 - Deliver an executive report detailing identified vulnerability vectors, recommended drift thresholds (typically 4 Nitro blocks ~1s), and recommended HAProxy stick-table rate limits.

4. SOP-02: Sidecar Deployment & HA Gateway Injection

Objective: Deploy DriftGuard sidecar instances in front of client workloads with zero client code changes.

```
Client Environment:  
[ Client Relay / App ] ===== HTTP POST (localhost:8545) =====> [ DriftGuard Sidecar Container ]  
    • HAProxy L7 Gateway  
    • Python 3.12 Sentinel Daemon  
    • Dynamic UNIX Socket Drain  
  
                                |                               |  
(Primary Healthy)|                                           |(Primary  
Drained)  
  
                                v                               v  
[ Client Validator Node ]   [ Private Fallback  
RPC ]
```

Deployment Procedures:

- **Option A (Kubernetes Sidecar Pattern):** Inject DriftGuard sidecar container into client's relayer or indexer pod spec using the official Helm chart. Set `RPC_URL=http://localhost:8545`.
- **Option B (Docker Compose / Systemd on Validator VPS):** Deploy DriftGuard directly on client's validator node VPS listening on `127.0.0.1:8545`.
- **UNIX Socket Verification Runbook:**

```
# Verify local HAProxy administration socket connectivity
echo "show stat" | socat stdio unix-connect:/run/haproxy/admin.sock | grep -E 'be_arb|be_nova'

# Verify local consensus sentinel healthz
curl -s http://localhost:8545/healthz | jq .
```

- **Zero-Downtime Traffic Canary Cutover:** Direct 10% traffic canary *to* verify zero errors over 1 hour *to* execute 100% cutover.

5. SOP-03: Consensus Anchor Calibration & Drift Tuning

Objective: Calibrate drift thresholds to prevent false positives during transient public internet jitter while ensuring sub-second cutover during real stalls.

Parameter	Default (Standard)	High-Velocity Orbit L3	High-Volume Relayer	Operational Purpose
poll_interval	0.2s (200ms)	0.1s (100ms)	0.15s (150ms)	Out-of-band polling loop frequency.
drift_threshold	4 blocks (1.0s)	3 blocks (750ms)	2 blocks (500ms)	Maximum allowed block divergence.
failure_threshold	1 poll	1 poll	1 poll	Consecutive failures before server drain.
recovery_threshold	2 polls	3 polls	3 polls	Consecutive healthy polls before un-draining.
syncing_check	true	true	true	Failover immediately if eth_syncing = true.

Flap Prevention Runbook: If public network jitter causes rapid flapping between primary and fallback, increase recovery_threshold to 3 or 4. Never decrease recovery_threshold below 2 to maintain hysteresis stability.

6. SOP-04: 24/7 Incident Escalation & Response Runbook

Objective: Handle upstream provider desyncs, sequencer stalls, and network partitions within strict contractual SLAs.

Severity Level	Definition	Response SLA	Target Resolution	Primary Actions
SEV-1 (Critical)	Primary & fallback both de-synced; client experiencing >0.1% transaction reverts; sequencer frozen.	< 15 minutes (Tier 2/3 on-call)	< 30 minutes	Automated failover verification; manual tertiary upstream injection via admin socket; client executive bridge opened.
SEV-2 (Major)	Primary upstream stalled; DriftGuard executed autonomous sub-130ms failover to fallback; client unaffected.	< 1 hour	< 2 hours	Verify fallback capacity; triage delinquent primary; notify upstream provider NOC; monitor for canonical recovery.
SEV-3 (Minor)	Transient consensus jitter (<2 blocks); single poll timeout; telemetry warning.	< 8 hours	Next business day	Review Redis telemetry logs; calibrate timeout thresholds if jitter persists.

SEV-1 Incident Escalation Workflow

- Trigger:** Sentinel trips critical alert to PagerDuty on-call paged (<60s) to Automated Discord/Telegram alert dispatched (<200ms).
- Diagnostics (<15 min):** Systems engineer inspects `curl -s http://localhost:8545/healthz` and verifies HAProxy state via admin socket.
- Fallback Verification:** Confirm autonomous socket drain successful and client HTTP 200 responses preserved.
- Emergency Tertiary Injection:** If all upstreams desync, inject emergency tertiary RPC into HAProxy pool at runtime via UNIX socket:

```
echo "set server be_arb/srv_fallback addr <new_ip> port 8545" | socat stdio unix-connect:/run/haproxy/admin.sock
```

- Post-Mortem:** Deliver formal post-mortem report to client within 24 hours.

7. SOP-05: Chaos Drills, Upstream Benchmarking & Monthly Reporting

Objective: Proactively stress-test the client's architecture and provide empirical scorecards on their upstream RPC vendors.

Bi-Weekly Chaos Drills (Tier 2 & 3)

- Conduct synthetic fault injections during scheduled maintenance windows:
 - ▶ Inject 20-block head freeze into primary RPC via `driftguard-chaos`.
 - ▶ Verify HAProxy drains primary within < **130ms**.
 - ▶ Measure packet drop rate (must remain **0.00%**).
 - ▶ Inject latency spike (>1000ms) to test hysteresis stability.

Monthly Upstream Vendor Scorecard

Every 30 days, DriftGuard delivers an executive resilience audit covering:

- **Uptime & Staleness Hours:** Total minutes each upstream provider lagged behind canonical head.
- **Average Drift-to-Drain Latency:** Measured failover speed across all recorded events.
- **Provider Reliability Rating:** Empirical ranking of client's vendors (Alchemy vs QuickNode vs Private Node).
- **Cost Optimization Advice:** Recommendations on reducing paid RPC tiers based on actual failover utilization.

8. SOP-06: Client Service Level Agreement (SLA) & Contractual Terms

1. Gateway Availability Guarantee (99.99%)

- DriftGuard guarantees **99.99% ingress gateway availability** for client workloads on Tier 2 and Tier 3 retainers.
- If availability falls below 99.99% in a monthly billing cycle, the client receives contractual service credits:
 - ▶ **99.90% – 99.98%:** 15% monthly retainer credit.
 - ▶ **99.00% – 99.89%:** 30% monthly retainer credit.
 - ▶ **< 99.00%:** 50% monthly retainer credit.

2. Data Privacy & Zero Key Custody

- DriftGuard engineers **never** take custody of private keys, mnemonic phrases, or transaction signing credentials.
- All RPC endpoints and consensus anchor credentials are encrypted via client-managed secret vaults or environment files.

3. Contractual Termination & Knowledge Transfer

- Retainer contracts operate on flexible month-to-month or quarterly commitments.
- Upon termination, DriftGuard delivers a complete infrastructure handoff pack, ensuring the client can continue running the open-source sidecar independently without vendor lock-in.

DRIFTGUARD SYSTEMS • ENTERPRISE OPERATIONS & B2B CLIENT RETAINER SOP • VERSION 2.0
Operations Desk: operations@driftguard.live • Emergency On-Call: PagerDuty / Telegram Priority Desk
Repository: github.com/maskalfreeup-glitch/driftguard • Portal: driftguard.live